

The arc is fine above 4 kHz. Below 2 kHz the room bends it by up to 8 dB.

We turned the propeller so that its axis points across the arc. Then every microphone sees the propeller from the same angle, and a perfect setup reads the same level at every position: a circle. We did this twelve times (keys prop7-18, 1-2 Sept 2026, PWM 1900). What is left after we remove each microphone's own offset is the effect of the room.

2026-09-08
data: laptop backup of 2 Sept
script: scripts/arc_error_map.py
full guide: docs/arc-validation-guide.html

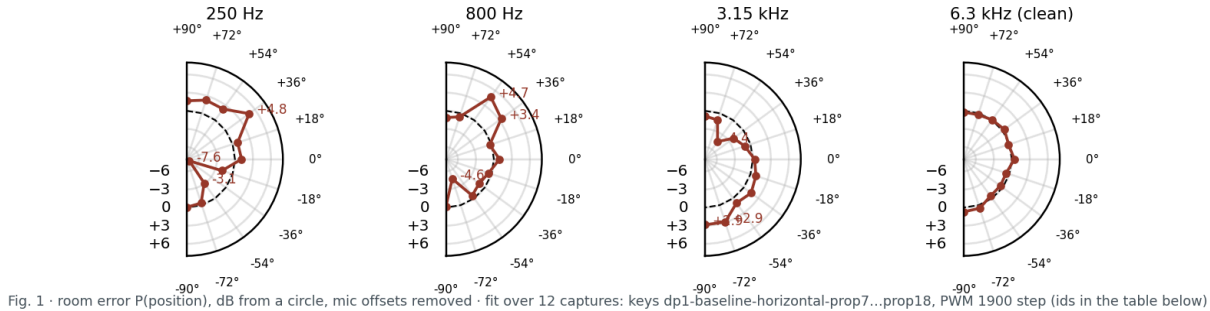


Fig. 1 · room error P(position), dB from a circle, mic offsets removed · fit over 12 captures: keys dp1-baseline-horizontal-prop7...prop18, PWM 1900 step (ids in the table below)

What the room does

The level at -72° is 2.4 to 5.7 dB too low between 500 and 2000 Hz, in all twelve runs. The dip is deepest at 630 and 800 Hz (-4.5 dB). It stays when a different microphone sits there (811-1896 in eight runs, 810-8901 in the four runs with the arc turned around). This position is the one closest to the chamber corner.

The level at $+36^\circ$ and $+54^\circ$ is 3 to 7 dB too high, at 200-250 Hz and at 630-1000 Hz. Three different microphones sat at $+54^\circ$ over the twelve runs. The bump stayed the same.

Around the blade tone (200-250 Hz) the two halves of the arc disagree: -36° reads -7.6 dB, $+36^\circ$ reads $+4.8$ dB; -54° reads -3 dB, $+54^\circ$ and $+72^\circ$ read $+2$ dB. The lower half of the arc (the microphones that look up at the propeller from the floor side) is too low, the upper half is too high. The dip also moves with motor speed (-9 dB at PWM 1900, -2 dB at 2000). A sound wave that bounces off a surface and adds to the direct wave does exactly this: it cancels at some frequencies and adds at others, and the tone slides through those frequencies as the speed changes.

Above 4 kHz the arc reads a circle within ± 1 dB at every position. The positions 0° , $\pm 18^\circ$ and $\pm 90^\circ$ are within about 1 dB in every band.

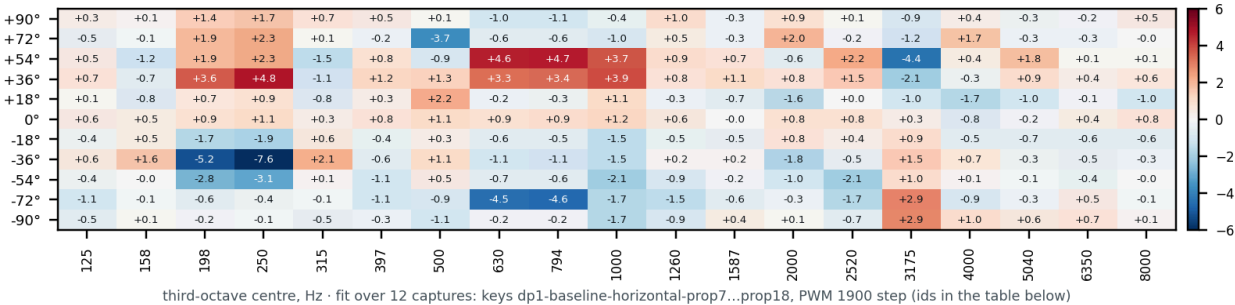
What it is not

Not the microphones, with one exception. Because the microphones at $+72^\circ$ and $+54^\circ$ were swapped after prop10, and the arc was turned around for prop11-14, we can tell apart what belongs to a microphone and what belongs to a position. Ten microphones agree within ± 1.8 dB in every band. **Microphone 811-1892 reads 2-6 dB too low from 125 Hz to 2.5 kHz.** Its calibration file does not match the capsule (its sensitivity value, -10.63 dB, is the least negative of the set, while the raw capsule is only about 1 dB low). When it sat at $+54^\circ$ it made the room error look like 7 dB. Take it off the arc or measure it again.

Not a labelling error. prop11, 12, 13 and 14 were measured with the arc turned around. Their patterns match the mirror image of the other runs (correlation $+0.79$ to $+0.98$). Plotted by label, the dip appears on the other side. Plotted by physical position, they agree with the other eight runs.

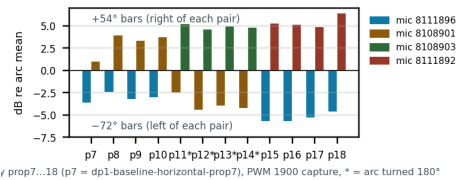
Not one frequency. The error sits in three separate bands, 200-250 Hz, 630-1000 Hz and 3150 Hz, and nowhere else (Fig. 2). Rajmane and Baumann (DAGA 2016) put steel plates into a good chamber and found that a plate only shows up once the wavelength is shorter than the plate: a 35 cm plate from 1.25 kHz, a 10 cm plate from 3.15 kHz. So three bands point to three objects of different size: about 1.5 m (floor, table or wall), about 0.4 m (stand base or a bar of the arc frame), about 0.1 m (a microphone clamp or mount).

Fig. 2 · Room error at each position and frequency band, in dB from a circle (microphone offsets removed; run-to-run scatter 0.6-2.3 dB)



third-octave centre, Hz · fit over 12 captures: keys dp1-baseline-horizontal-prop7...prop18, PWM 1900 step (ids in the table below)

Fig. 3 · Two positions, twelve runs, three microphones



500-2000 Hz, microphone offsets removed; * = arc turned around. The dip at -72° and the bump at $+54^\circ$ keep their size no matter which microphone sits there.

Measurements plotted (all figures): the last PWM-1900 capture in each key 2004_6in_unset_dp1-baseline-horizontal-<key>, 11 microphones $\times$ 2.0 s, calibrated; microphone serials 810-/811-xxxx given by their last four digits

KEY	CAPTURE (PWM 1900)	ARC	+72/+54
prop7	2026-09-01 16:42:31	norm	1892/8901
prop8	2026-09-01 16:44:39	norm	1892/8901
prop9	2026-09-01 16:50:45	norm	1892/8901
prop10	2026-09-01 16:55:35	norm	1892/8901

KEY	CAPTURE (PWM 1900)	ARC	+72/+54
prop11	2026-09-02 13:18:49	180°	8901/1892
prop12	2026-09-02 14:24:12	180°	8901/1892
prop13	2026-09-02 14:34:37	180°	8901/1892
prop14	2026-09-02 14:37:02	180°	8901/1892

KEY	CAPTURE (PWM 1900)	ARC	+72/+54
prop15	2026-09-02 15:48:59	norm	8901/1892
prop16	2026-09-02 15:54:26	norm	8901/1892
prop17	2026-09-02 16:52:44	norm	8901/1892
prop18	2026-09-02 17:08:19	norm	8901/1892

Left out of every figure: the 31 Aug keys (...-2026-08-31, -2,...-6) and prop1-prop6 (1 Sept morning). Their patterns match neither arc orientation, so the rig or the room was different then. "180°" means the label +e was physically at -e; those four runs are mirrored before plotting. The offset of microphone 811-1892 is removed everywhere. Figs. 1-2 show the fitted position term P; Fig. 3 shows single runs.

Method: Welch spectrum (4096-point Hann), UMIK-2 calibration as the app applies it, third-octave sums from 125 Hz to 8 kHz. Model: level = run gain + position term P + microphone term M, fitted by least squares over prop7-18 with prop11-14 mirrored. Every number comes from the files named above; CSV tables in docs/analysis/. Literature: papers/small-chamber/ and papers/chamber-problems/.